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And

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1. INTRODUCTION

UKF Company operates trading floors across various locations in the UK, depending significantly on a resilient and high-performing network infrastructure to facilitate crucial trading activities. However, the prevailing network infrastructure is no longer adequate to fulfil the strict requirements for availability and performance that are indispensable in today's swiftly changing trading landscape. The current hardware and cabling elements are antiquated and necessitate immediate enhancement, in conjunction with a thorough overhaul of the network framework. In light of these obstacles, we propose a resolution that not only tackles the present shortcomings but also ensures future-proofing for the subsequent 5 to 10 years. Our aim is to provide an innovative network infrastructure that not only satisfies but surpasses UKF's operational needs, guaranteeing continuous service levels and bolstering the company's expansion and prosperity well into the future.

2. BACKGROUND

UKF Company is positioned as a leading financial entity within the United Kingdom, renowned for its significant contribution to the financial domain. The organization possesses and manages two prominent trading venues located in London, which serve as focal points for financial dealings and market operations. Strategically situated in central London, our Head Office supervises our nationwide endeavours and strategic ventures.

Beyond our primary location, UKF Company upholds regional establishments in pivotal urban canters such as Birmingham, Manchester, and Edinburgh. These regional branches play a vital role in expanding our financial offerings to enterprises and sectors throughout Britain, thereby enhancing economic development and resilience across diverse fields.

UKF remains steadfast in its commitment to providing comprehensive financial services customized to meet the distinct requirements of our clientele. Drawing upon our extensive knowledge and network, we strive to assist businesses of varying scales and industries across the United Kingdom. Our unwavering dedication to excellence and ingenuity solidifies our standing as a dependable ally within the British financial sphere.

TECHNICAL FRAMEWORK

- The operating system in use on desktop computers is currently Windows.
- Servers exhibit a combination of both Windows and Linux operating systems.
- Network communication protocol is TCP/IP, specifically utilizing IPv4 at the present time.
- Employed Layer 2 switches along with routers. - The Open Shortest Path First (OSPF) protocol is utilized for internal routing purpose.

3. PURPOSE OF PROPOSAL

- Optimal Performance
- Enhanced Availability
- Scalability, supporting diverse devices including mobile platforms.
- Ease of management, ensuring cost-effective operations and reduced Total Cost of Ownership (TCO) compared to our current network.

4. OBJECTIVES OF PROJECT

The aim of this project is to devise and execute a novel network infrastructure for the 2nd trading floor in London and to establish a resilient Wide Area Network (WAN) connection that links all regional offices of our organization. The project's scope encompasses the network services necessary for the trading floor, catering to a total of 500 traders, in addition to the essential services for the support office, accommodating 45 support staff and a management team consisting of 15 individuals.

The specific goals includes:

a. Provision of network connectivity and services to facilitate the trading activities of 500 traders during core hours from 6am to 6pm.
b. Ensuring smooth communication and network resource access for the support staff and management team across both the trading floor and the support office.
c. Provision of desktop PCs for all staff members, including sufficient networked printers strategically placed in designated areas.

The objective is to boost operational efficiency, facilitate business expansion, and establish a network foundation that is future-proof and aligns with the evolving demands of our organization.

5. KEY REQUIREMENTS

NETWORK PERFORMANCE OPTIMIZATION

The task at hand involves the development and execution of an innovative network infrastructure for the secondary trading area located in London, in conjunction with a resilient Wide Area Network (WAN) connection that interconnects our various regional offices. The primary objective of this endeavour is to bolster the trading operations of 500 traders during peak hours, guaranteeing smooth communication channels and network connectivity for a support team of 45 individuals and a managerial cohort comprising 15 members, spanning both the trading floor and support premise. Our aim is to equip all personnel with desktop computers while strategically placing networked printers to streamline operational processes. The primary emphasis of this endeavour is on elevating operational efficacy, facilitating business growth, and establishing a resilient network framework that is in harmony with our ever-evolving organizational requirements. Furthermore, strict performance benchmarks dictate that all trading transactions on the trading floor must be finalized in under 1ms, with ancillary transactions wrapping up in less than 5ms, thereby highlighting the paramount significance of optimizing network performance.

5.1 AVAILABILITY

Network availability is main role for maintaining continuous access to services and resources, representing the duration a network remains functional. Achieving a high level of availability requires incorporating redundancy, fault tolerance, proactive monitoring, and disaster recovery strategies. Redundancy ensures backup elements are ready to assume responsibilities in case of malfunctions. Proactive monitoring facilitates early issue detection, while disaster recovery plans ensure swift restoration after disturbances. Collectively, these approaches uphold dependable network services, crucial for contemporary business operations and interpersonal exchanges.

This initiative aims to enhance the network infrastructure for the organization by ensuring high availability. Specifically, the trading floor must have a 99.999% availability rate, while support services should maintain a 99% availability rate.

5.2 FLEXIBILITY

As networks operate as a fundamental social framework, the maintenance of stability becomes a crucial need. Moreover, it is imperative to incorporate the capability to adapt to the aforementioned alterations. The notion of flexibility emerges as a pivotal objective in the evolution of networks. This adaptability is facilitated through the virtualization of networking and computational resources, encompassing bandwidth, routing functionalities, servers, and storage, which are then amalgamated dynamically to innovate new services, optimize resource distribution, mitigate system failures, and other related functions. we will provide assistance to enable the network to accommodate wireless mobile devices, albeit not meeting identical availability criteria.

5.3 FUTURE PROOF

To guarantee the prosperity of your network infrastructure in the forthcoming five years, it is crucial to concentrate on allocating resources towards cutting-edge technologies such as SDN and continual performance monitoring. It is essential to keep redundancy protocols current to ensure seamless connectivity and consistently assess contingency plans for emergencies. Maintaining an adaptable stance to accommodate evolving business demands, giving precedence to security, and dedicating resources to employee training are imperative for efficaciously managing the network. By adopting a proactive strategy towards administration and enhancement, your network will effectively bolster the secondary trading sector in London with robustness and effectiveness.

5.4 SECURITY

Data protection is an important role for the enterprise. The repercussions of data loss stemming from malware or phishing attacks can result in significant setbacks for the organization, with potential consequences as dire as business failure for smaller enterprises. The implementation of robust network security measures serves as a shield against such losses, effectively safeguarding both business and personal data. The effectiveness and efficiency of a secure network stem from the absence of malevolent software infiltrations.

A combination of network security protocols and a well-structured system contribute to enhancing network performance, ensuring seamless operations and optimal functionality. In the absence of adequate network security measures, cybercriminals can exploit vulnerabilities to steal your organization's brand identity.

Through unauthorized access to your systems, these malicious actors can misuse your identity for fraudulent purposes. Mitigating such risks necessitates the establishment of robust network security protocols to protect your business interests. Employ firewalls and updated high-level security protocols to mitigate unauthorized access. Keep security patches updated and use web filtering for protection Secure connections for remote workers by employing VPN technologies with strong encryption and multi-factor authentication. Enforce strict access controls and monitor activities for suspicious behaviour Implement internal-security mechanisms like robust access controls, authentication methods, and network segmentation. Regularly audit access permissions to mitigate insider threats Enhance network security by implementing these measures against internal and external threats.

6. FUNCTIONAL REQUIREMENTS

NETWORK DESIGN

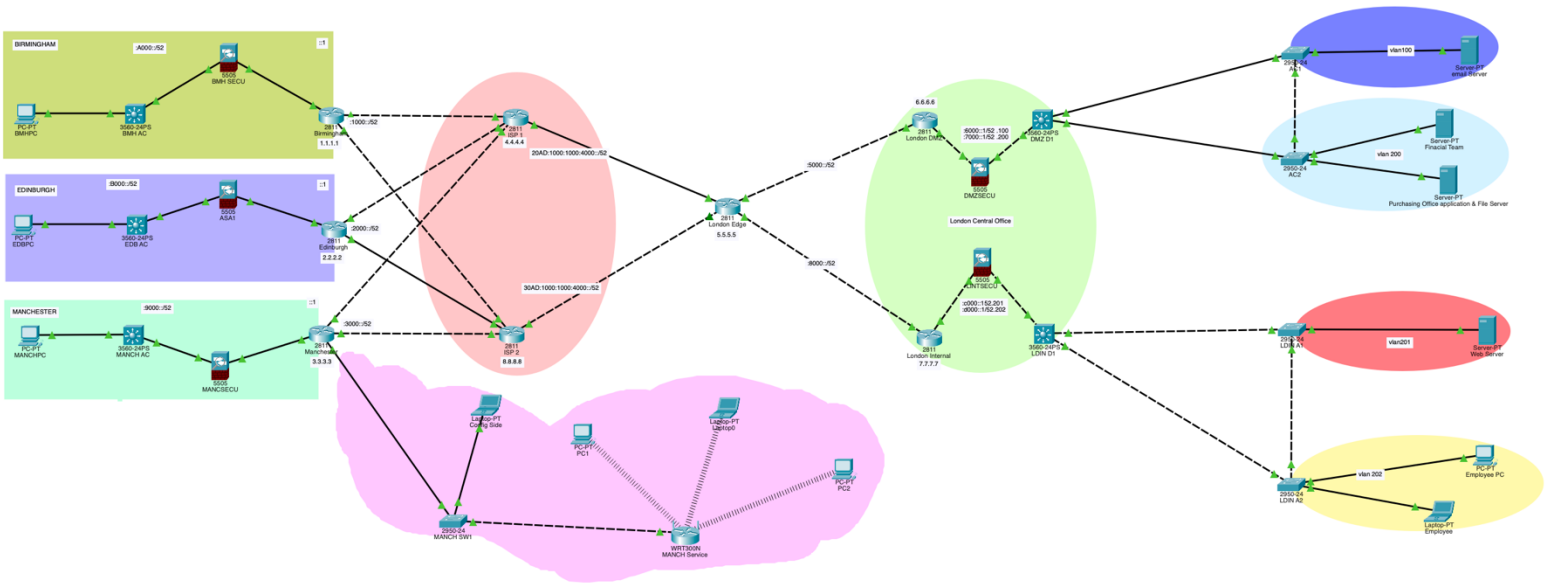


Fig 1: Potential Network Design

Our design approach will give precedence to incorporating sophisticated technologies in order to satisfy the strict performance and availability criteria for trader data. Moreover, we will integrate resilient redundancy and fault tolerance mechanisms to secure uninterrupted connectivity for crucial trading activities. While guaranteeing optimal performance for trader data, we will structure the network to support other types of traffic with less demanding prerequisites, thereby maximizing resource utilization and cost efficiency. To cater to your company's requirement for a network solution that guarantees consistent service for a minimum of 5 years, we will utilize scalable and future-proof hardware and software elements. This approach ensures adaptability to emerging technology trends and business demands, ensuring long-lasting sustainability. In accordance with your objective of diminishing overall power consumption by 25% within the upcoming 10 years, our proposal will encompass energy-efficient hardware and smart power management techniques. This will lead to notable enhancements in power consumption, fostering sustainability and financial savings.

we employed software known as Packet Tracer to analyse the potential network infrastructure, encompassing network components such as routers, switches, firewalls, PCs, servers, and wireless routers. The linkage between two primary regions, one comprising the Edinburgh, Birmingham, and Manchester regions and the other being the London internal region, has been transitioned. Two distinct Internet service providers were utilized, with firewalls positioned between the primary source routers and main switches to enhance security measures. A trunk link was established to augment redundancy along associated paths. The utilization of two internet service providers further bolsters redundancy within this network connection. Moreover, essential components such as an email server, a web server, and DNS have been incorporated. It is also plausible to operate various servers, including a DHCP server, file server, and HTTP server, in a secluded manner within this network setup. This serves as a prototype design for the infrastructure, bearing in mind that the actual implementation in a real-world scenario would be notably intricate, necessitating adaptations across all layers of network services.

7. PHYSICAL LAYER

7.1 NETWORK DEVICES

Routers

Routers serve the purpose of connecting computers and various other devices to the vast network that is the Internet. Functioning as a dispatcher, a router meticulously selects the most optimal path for the transmission of your data. Moreover, routers possess the capability to determine the precedence of certain computers over others in terms of network access.

we employed corresponding routers as the primary means of connectivity across all regions. Utilizing routers facilitates the connection of numerous devices to the Internet and establishes connections between said devices. Furthermore, routers can be utilized to establish local networks of devices, which prove beneficial for file sharing among devices or enabling employees to collaborate on software applications.

The absence of routers would result in the misdirection of data within your business operations. For instance, when desiring to print a document, a router becomes essential in ensuring the document is directed to the printer, as opposed to another computer or scanner.

Switches

Switches play a crucial role as fundamental components in any network infrastructure. They establish connections among numerous devices, including computers, wireless access points, printers, and servers, all operating within a common network in a physical location like a building or campus. By facilitating communication, a switch empowers the interconnected devices to exchange data and engage in interactions.

Cables

In the realm of networking's physical layer, the pivotal role is assumed by the connecting cable. Presently, optical fibre cables, including single mode and duplex mode such as APC and UPC, are extensively employed. The selection of cable type is contingent upon the specific circumstances; for instance, Coaxial cable is ideal for short-length physical device connections, while RJ45 cables are predominantly utilized for linking PCs, switches, and routers within close proximity. The choice of cables varies depending on the distance of the connectivity.

7.2 NETWORK MAPPING

Networking cartography is a potent instrument for network administrators. Such cartographic representations aid enterprises in acquiring profound understanding of a network's functionality and configuration. Diverse varieties of network maps are employed to depict individual networks.

Physical Network Map

This cartographic representation depicts the tangible stratum of a network, commonly known as a "rack diagram". Physical network mappings generally delineate the geographical placement of network apparatus, their interconnections, and the nature of the links linking them.

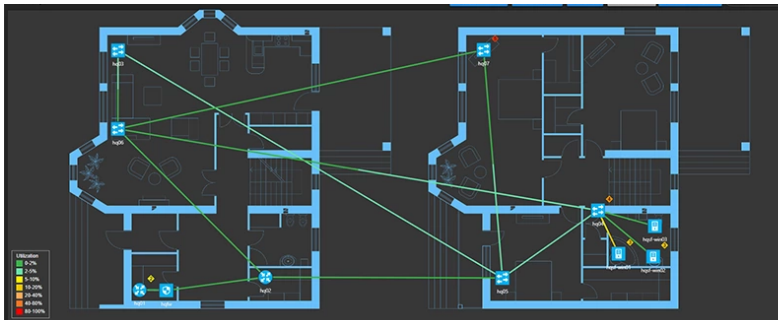


Fig 2: Physical Network Map (Example)

We shall construct precise physical network diagrams for every level of the corporate premises subsequent to conducting an analysis of the architectural layout of the company edifice. Moreover, we intend to furnish an immaculate allocation of frequencies for each wireless device and ensure minimal attenuation of data traffic passing through the cables. This encompasses activities such as overseeing the installation of cables, determining optimal placements for wireless routing devices, managing access points, assessing the thickness of the building structure, and balancing the bandwidth distribution.

Layer 2 Network Map

Layer 2 Network Maps will illustrate the presence of equipment such as switches operating within the confines of layer 2 protocol specifications such as the Spanning Tree Protocol (STP). The primary advantage of utilizing a Layer 2 map is evident when one endeavours to trace the trajectory of data flow depicted in a diagram, enabling a clear understanding of the direction in which traffic navigates based on its spanning tree priority.

Layer 3 Network Map

Layer 3 Network Maps comprise elements such as routers, firewalls, and load balancers, in addition to IP addressing, route details, and layer segments extending from each network interconnect. The advantage lies in the ability to trace a data trajectory from its origin to its endpoint and discern the route it would traverse. This capability proves indispensable in troubleshooting scenarios.



Fig 3: Connectivity in UK

8. DATALINK LAYER

Data link protocols such as Ethernet for local area networks (multi-node), the Point-to-Point Protocol (PPP), HDLC, and ADCCP for point-to-point (dual node) connections are commonly used. Within the Internet Protocol Suite (TCP/IP), the functionality of the data link layer is encompassed within the link layer, which represents the lowest layer in the descriptive model.

8.1 High-level Data Link Control (HDLC)

It comprises a set of communication protocols situated at the data link layer, facilitating data transmission among network points or nodes. Due to its nature as a data link protocol, information is structured into frames, which are subsequently sent across the network to the intended recipient for validation of their proper receipt. Operating as a protocol oriented towards bits, it is versatile in its application, catering to both point-to-point and multipoint communication scenarios.

8.2 Spanning Tree Protocols

The main objective of STP is to solve the challenges surrounding data transmission in a local area network (LAN) with redundant pathways. In the absence of meticulous traffic management, data packets may traverse a cyclic route across network segments, thereby impeding network performance and causing congestion.

Network configurations frequently incorporate redundant pathways to interconnect various segments. While redundancy serves as a safeguard against failures, it also has the potential to trigger bridge or switch looping. This phenomenon occurs when data packets follow multiple paths from the source to the destination, leading to a cyclical propagation of data along these pathways, consequently intensifying the traffic volume and culminating in a broadcast storm.

STP plays a crucial role in averting bridge looping within LANs that incorporate redundant connections. The absence of STP would pose challenges in leveraging redundancy without risking network looping. By scrutinizing all network links, STP can pinpoint redundant links and selectively deactivate ports that might instigate looping behaviours.

8.3 Cisco Discovery Protocol (CDP)

Network protocol at layer 2 developed by Cisco is utilized as a tool for network discovery, automatically activated by default on Cisco devices. CDP functions on Cisco devices to detect other Cisco devices present in the network. Through the utilization of CDP, individuals can thoroughly analyse devices without necessitating physical examination. This protocol provides individuals with the capability to utilize specific show commands to access information regarding connected devices, such as details on local and remote ports, hostnames, device platforms, and additional data.

The utilization of RTP (Real-Time Transport Protocol) and various network-layer protocols is permitted for the purpose of identifying devices and detailing their configurations. Assistance is provided in resolving issues related to TLV (Type Length Value) fields. It serves as a diagnostic instrument to aid in the resolution of problems concerning devices and networks. The identification of the IP address of a misconfigured switch/router across a WAN-link is facilitated.

9. NETWORK LAYER

The network layer is accountable for the establishment of data routing paths to enable network communication, with data being sent to the receiving device in packet form via logically defined network routes in a sequential fashion overseen by the network layer. Key responsibilities of the network layer consist of the establishment of logical connections, data forwarding, routing, and error reporting in delivery. Layer 3 possesses the capacity to back either connection-oriented or connectionless networks, albeit not concurrently.

9.1 Internet Protocol Security (IPsec)

internet Protocol Security, also known as IPsec, is a comprehensive security protocol suite designed to safeguard data from unauthorized disclosure during transmission between storage systems and clients. By implementing IPsec, one can establish specific policies on the storage system to configure encryption and authentication algorithms for communication with clients. This enables the negotiation of a security association (SA) between the two end-stations involved in initiating and receiving secure communications. The utilization of this security association is crucial for facilitating secure data exchanges between the storage system and the client systems.

9.2 Virtual Private Network (VPN)

A Virtual Private Network (VPN) is a technological tool enabling individuals to establish a secure and encrypted link between two distant computers through the utilization of the internet. The provision of a secure connection by a VPN service assists users in gaining access to potentially restricted resources. The limitations on accessing these resources may stem from various reasons such as geographical location, network regulations, or other relevant constraints.

10. FINACIAL MANAGEMENT & COST EFFECTIVE

we are planning to provide these specific features in order to effectively minimize the overall expenses associated with the implementation of this particular network solution. This strategic approach aims to optimize cost-efficiency and streamline the financial resources required for the successful deployment and maintenance of the network infrastructure.

10.1 Network Segmentation

Network segmentation is a cost-effective strategy that improves security and operational effectiveness in a network infrastructure. Through the partitioning of the network into smaller, isolated sections, organizations can manage the access to sensitive data and resources more efficiently, thereby decreasing the likelihood of unauthorized access and data breaches. This method limits the repercussions of security incidents by confining threats within specific segments, consequently reducing potential harm and the associated costs of remediation. Furthermore, network segmentation enhances resource distribution by guiding traffic in a more effective manner, enhancing overall network performance and diminishing bandwidth congestion. In general, the implementation of network segmentation provides a cost-effective means to enhance security, streamline operations, and mitigate risks, establishing it as a crucial element of any comprehensive network solution.

10.2 Out Source Non-Core Functions

Outsourcing peripheral functions constitutes a strategic manoeuvre embraced by numerous enterprises to rationalize operations and concentrate resources on essential undertakings. Delegating non-essential tasks, such as technical support, client assistance, or salary processing, to external service providers enables establishments to leverage specialized proficiency, heightened efficacy, and financial advantages. The

practice of outsourcing empowers firms to tap into a worldwide talent pool and access top-tier solutions devoid of substantial investments in infrastructure or personnel. Furthermore, outsourcing allows for scalability, enabling enterprises to calibrate resources in accordance with variable demands and market circumstances. This adaptability facilitates prompt adjustments to evolving business environments and more strategic allocation of resources. Moreover, outsourcing peripheral functions liberates internal resources, affording employees the opportunity to focus on fundamental competencies and strategic endeavour that propel business expansion and ingenuity.

10.3 Implement Virtualization

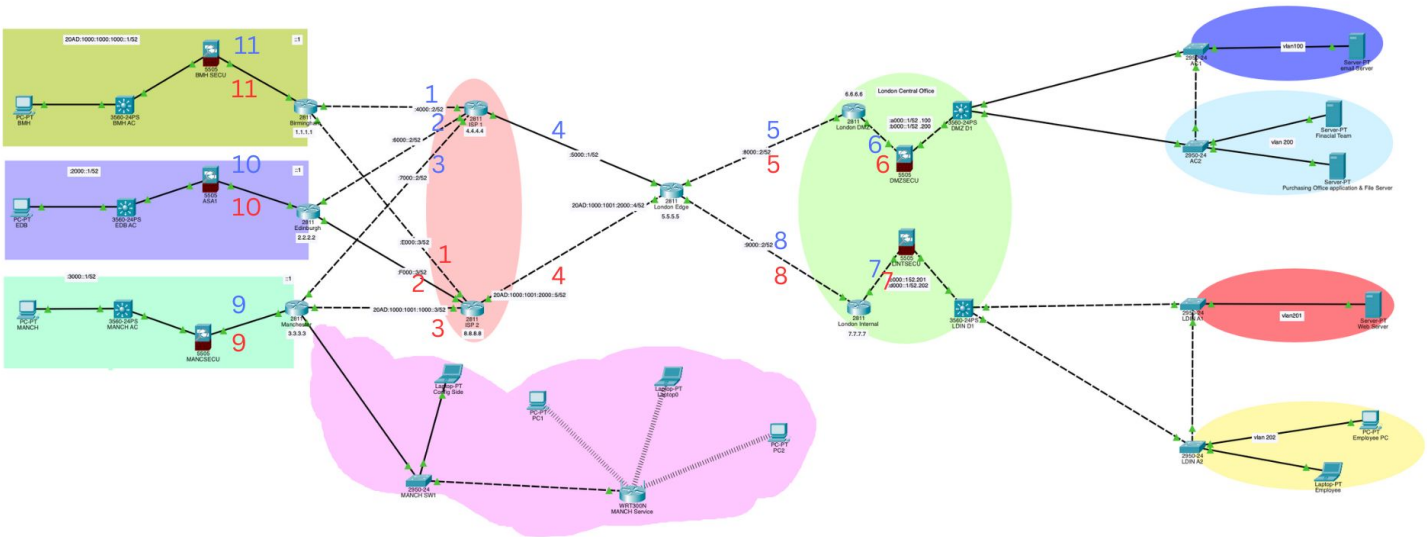
Implementing virtualization in network deployment offers a plethora of cost-effective advantages for organizations. Embracing virtualization technologies can lead to a substantial reduction in hardware needs, resulting in decreased initial investments and continuous operational costs. Virtualization permits the amalgamation of numerous virtual instances onto a singular physical device, optimizing resource utilization and minimizing the spatial requirements of network infrastructure. This amalgamation not only cuts down on hardware expenses but also diminishes energy consumption and associated maintenance costs. And then ,virtualizing network functions like firewalls, routers, and switches provides enhanced flexibility in resource allocation, enabling organizations to adapt more effectively to evolving business demands without expensive hardware upgrades. Additionally, virtualized environments simplify network administration through centralized management and automated provision of network services, decreasing administrative burdens and streamlining operations. As underscored in industry literature, the incorporation of network virtualization presents a strategic opportunity for firms to optimize expenses, boost scalability, and foster innovation in network infrastructure deployment. By proficiently implementing virtualization, organizations can realize substantial cost reductions while enhancing agility, efficiency, and competitiveness in the current dynamic business environment.

10.4 Network Implementation

Through the implementation of automated and orchestrated infrastructures that eliminate laborious manual tasks, organizations can not only enhance the capabilities of their networks but also realize a quicker return on investment. The utilization of automation also plays a crucial role in minimizing errors and enhancing the resilience of the network. Many automation solutions are designed to automatically address

network errors without requiring manual intervention, thereby bolstering business resilience and ensuring seamless access to necessary applications and data for employees.

11. Ipv6 address Table & IP Scheme



IPV6 (Internet Protocol Version 6)

IPv6 serves as the most recent iteration of the Internet Protocol, serving the function of identifying various devices dispersed throughout the expanse of the internet, thereby enabling their precise location. The assignment of a unique IP address to each device utilizing the internet is imperative to facilitate seamless communication over the network. Analogous to the significance of street addresses and zip codes requisite for the successful delivery of mail, IP addresses play a crucial role in enabling effective internet communication.

Subnetting

The necessity for the application of the subnetting concept lies in its ability to offer an alternative method for mitigating IP address wastage. Subnetting involves the partitioning of an IP address into numerous smaller components, which function as distinct network entities referred to as subnets within the host network. This process of subnetting enables the network to circumvent frequent congestion issues within the channel, thereby facilitating seamless transmission of data.

Address Tables for ISP 1 (Multi-homing)

Level 1 Address Block (ISP 1 Prefix)

Subnet No.	Traditional Subnetting	Symbolic Subnetting	
1	20AD:1000:1000:0000::/52	20AD:1000:1000:0000::/52	Infrastructure
2	20AD:1000:1000:2000::/52	20AD:1000:1000:1000::/52	User
3	20AD:1000:1000:4000::/52	20AD:1000:1000:2000::/52	Reserved

Level 2 Address Block for infrastructure (ISP 1 Subnet 1 Prefix)

Index	Infrastructure 20AD:1000:1000:0000::/52	Assign to
	Symbolic Subnetting	

1	20AD:1000:1000:1000::/52	Birmingham – ISP 1
2	20AD:1000:1000:2000::/52	Edinburgh – ISP 1
3	20AD:1000:1000:3000::/52	Manchester – ISP 1
4	20AD:1000:1000:4000::/52	ISP 1 - Ledge
5	20AD:1000:1000:5000::/52	Ledge-LDMZ
6	20AD:1000:1000:6000::/52	LDMZ – Email Server
7	20AD:1000:1000:7000::/52	LDMZ – Financial Floor
8	20AD:1000:1000:8000::/52	Ledge - London Internal
9	20AD:1000:1000:9000::/52	Manchester Internal
10	20AD:1000:1000:B000::/52	Edinburgh Internal
11	20AD:1000:1000:A000::/52	Birmingham Internal

12	20AD:1000:1000:C000::/52	London Internal – Web Server
13	20AD:1000:1000:D000::/52	London internal – Employee PCs
14	20AD:1000:1000:E000::/52	FG 3
15	20AD:1000:1000:F000::/52	MANCH Internal – WIFI service

Level 3 Address Block for infrastructure (ISP 1 Subnet 1 Prefix)

Index	Infrastructure 20AD:1000:1000:0000::/52 Symbolic Subnetting	Assign to
1	20AD:1000:1000:1000::1/52 20AD:1000:1000:1000::2/52	ISP 1 Birmingham
2	20AD:1000:1000:2000::1/52 20AD:1000:1000:2000::2/52	ISP 1 Edinburgh

3	20AD:1000:1000:3000::1/52 20AD:1000:1000:3000::2/52	ISP 1 Manchester
4	20AD:1000:1000:4000::1/52 20AD:1000:1000:4000::2/52	ISP 1 Ledge
5	20AD:1000:1000:5000::/52 20AD:1000:1000:5000::/52	Ledge LDMZ
6	20AD:1000:1000:6000::1/52 20AD:1000:1000:6000::100/52	LDMZ Email Server
7	20AD:1000:1000:7000::1/52 20AD:1000:1000:7000::200/52 20AD:1000:1000:7000::203/52	London BR Internal Financial Team Server Sub-File Server
8	20AD:1000:1000:8000::1/52 20AD:1000:1000:8000::2/52	Ledge London internal
9	20AD:1000:1000:9000::1/52 20AD:1000:1000:9000::100/52	Manchester Internal MANCHPC

10	20AD:1000:1000:A000::1/52 20AD:1000:1000:A000::100/52	Birmingham Internal PC1
11	20AD:1000:1000:B000::1/52 20AD:1000:1000:B000::100/52	Edinburg Internal PC2
12	20AD:1000:1000:C000::1/52 20AD:1000:1000:C000::201/52	London Internal Web server
13	20AD:1000:1000:D000::1/52 20AD:1000:1000:D000::202/52 20AD:1000:1000:D000::204/52	London Internal (sec) Employee PC1 Employee PC2
14	20AD:1000:1000:E000::/52	FG3
15	20AD:1000:1000:F000::2/52 20AD:1000:1000:F000::100/52	MANCH Internal Wi-Fi service (Config Side)

Address Tables for ISP 2 (Multi-homing)

Level 1 Address Block (ISP 2 Prefix)

Subnet No.	Traditional Subnetting	Symbolic Subnetting	
1	30AD:1000:1000:0000::/52	20AD:1000:1000:0000::/52	Infrastructure
2	30AD:1000:1000:2000::/52	20AD:1000:1000:1000::/52	User
3	30AD:1000:1000:4000::/52	20AD:1000:1000:2000::/52	Reserved

Level 2 Address Block for infrastructure (ISP 2 Subnet 1 Prefix)

Index	Infrastructure 30AD:1000:1000:0000::/52	Assign to
	Symbolic Subnetting	
1	30AD:1000:1000:1000::/52	Birmingham – ISP 1

2	30AD:1000:1000:2000::/52	Edinburgh – ISP 1
3	30AD:1000:1000:3000::/52	Manchester – ISP 1
4	30AD:1000:1000:4000::/52	ISP 1 - Ledge
5	30AD:1000:1000:5000::/52	Ledge-LDMZ
6	30AD:1000:1000:6000::/52	LDMZ – Email Server
7	30AD:1000:1000:7000::/52	LDMZ – Financial Floor
8	30AD:1000:1000:8000::/52	Ledge - London Internal
9	30AD:1000:1000:9000::/52	Manchester Internal
10	30AD:1000:1000:B000::/52	Edinburgh Internal
11	30AD:1000:1000:A000::/52	Birmingham Internal
12	30AD:1000:1000:C000::/52	London Internal – Web Server
13	30AD:1000:1000:D000::/52	London internal – Employee PC

14	30AD:1000:1000:E000::/52	FG 3
15	30AD:1000:1000:F000::/52	MANCH Internal – WIFI service

Level 3 Address Block for infrastructure (ISP 2 Subnet 1 Prefix)

Index	Infrastructure 30AD:1000:1000:0000::/52 Symbolic Subnetting	Assign to
1	30AD:1000:1000:1000::1/52 20AD:1000:1000:1000::2/52	ISP 1 Birmingham
2	30AD:1000:1000:2000::1/52 30AD:1000:1000:2000::2/52	ISP 1 Edinburgh
3	30AD:1000:1000:3000::1/52 30AD:1000:1000:3000::2/52	ISP 1 Manchester

4	30AD:1000:1000:4000::1/52 30AD:1000:1000:4000::2/52	ISP 1 Ledge
5	30AD:1000:1000:5000::/52 30AD:1000:1000:5000::/52	Ledge LDMZ
6	30AD:1000:1000:6000::1/52 30AD:1000:1000:6000::100/52	LDMZ Email Server
7	30AD:1000:1000:7000::1/52 30AD:1000:1000:7000::200/52 30AD:1000:1000:7000::203/52	London BR Internal Financial Team Server Sub-File Server
8	30AD:1000:1000:8000::1/52 30AD:1000:1000:8000::2/52	Ledge London internal
9	30AD:1000:1000:9000::1/52 30AD:1000:1000:9000::100/52	Manchester Internal MANCHPC
10	30AD:1000:1000:A000::1/52 30AD:1000:1000:A000::100/52	Birmingham Internal PC1

11	30AD:1000:1000:B000::1/52 30AD:1000:1000:B000::100/52	Edinburg Internal PC2
12	30AD:1000:1000:C000::1/52 30AD:1000:1000:C000::201/52	London Internal Web server
13	30AD:1000:1000:D000::1/52 30AD:1000:1000:D000::202/52 30AD:1000:1000:D000::204/52	London Internal (sec) Employee PC1 Employee PC2
14	30AD:1000:1000:E000::/52	FG3
15	30AD:1000:1000:F000::2/52 30AD:1000:1000:F000::100/52	MANCH Internal Wi-Fi service (Config Side)

12. Conclusion

In summary, the attempt to create a resilient network infrastructure for the second trading floor located in London emerges as a crucial initiative aimed at meeting the diverse operational demands of a considerable workforce comprising 500 traders and support personnel. The primary goals of this undertaking revolve around augmenting operational efficacy by establishing a seamless mode of communication, ensuring dependable access to network resources, and introducing desktop PCs to streamline the facilitation of business expansion opportunities. The implementation of robust security protocols stands out as a pivotal aspect in preserving the integrity and confidentiality of data, encompassing strategies such as fortifying external internet connections, securing remote worker links, and integrating internal security measures within the network framework.

The ultimate success of this project hinges upon the ability to fulfil exacting criteria pertaining to performance, availability, adaptability, and security. Through the realization of these outlined objectives, an enduring foundation will be laid that not only caters to existing operational requirements but also positions the organization favourably for prospective expansion and technological progression. This sophisticated network infrastructure is poised to empower the entity in adapting to the dynamic shifts within the commercial domain and harnessing technological advancements, thereby ensuring a sustained competitive edge and a pinnacle of operational proficiency in the forthcoming years.

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